

FABRIZIO RUSSO

Research Fellow, Imperial College London

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Scholar

dblp

SUMMARY

With six years' experience consulting and leading data-science teams in banking, I bridge academia, industry and regulatory bodies to drive human-centred AI innovation. Building on my PhD thesis in Safe & Trusted AI, and my postdoctoral research on explainable and contestable causal discovery (CARLA), I am now a Research Fellow at Imperial, translating explainable, interactive causal discovery research into practical tools for finance, healthcare and policymaking. I've published at established peer-reviewed venues while building an independent academic profile aimed at, and grounded on, real-world impact.

WORK EXPERIENCE

Imperial College London

- Research Fellow (AI) Imperial Policy Forum Full-time: Sep 2026-present UK
- Research Associate Full-time: Mar 2025-Aug 2026 UK
 - ERC Proof of Concept grant: Causal Argumentative Learning Assistant (CARLA)
- Research Assistant Full-time: Jan 2025-Feb 2025 UK
- Graduate Teaching Assistant Part-time: Oct 2021 & 2022 UK

Consultant

- Imperial College Consultants (ICON) Part-time: Mar 2022 – Sep 2022 UK & USA
 - Research on outlier identification and forecast for UK commercial properties index
- Ernst & Young (in collaboration with STAI CDT) Part-time: Jan 2021 – Apr 2021 UK
 - Research on back-door poisoning attacks identification for Large Language Models

4most Europe Ltd

- Head of Data Science Feb 2018 – Sep 2020 UK & Germany
 - Responsible for the development and delivery of data science projects for UK and German financial institutions
 - Design, prototype and build an interactive ML model monitoring tool (Tech of the Year - National Credit Awards '22)
- Managing Consultant Jul 2016 – Jan 2018 UK & Canada
 - Responsible for statistical modelling and analysis of Credit Risk for UK and Canadian financial institutions
 - Responsible for the development and performance of a team of 3 junior consultants
- Credit Risk Consultant Jul 2015 – Jun 2016 UK

- Credit Risk Analyst Full-time: May 2014 – Jun 2015
- General Electric Capital - UK Home Lending UK

- Sales Assistant Apr 2013 – Apr 2014
- Privileged Ltd & Hollister UK

EDUCATION

- PhD in Safe and Trusted Artificial Intelligence Full-time: Oct 2020 – Feb 2025
- Imperial College London UK

- Thesis: Causal Discovery for Trustworthy Artificial Intelligence
- Funded by UKRI Safe and Trusted Artificial Intelligence Centre for Doctoral Training

- MSc in Applied Statistics (Distinction) Part-time: Oct 2016 – Oct 2018
- University of London - Birkbeck College UK

- Dissertation: Macroeconomic Modelling through VAR and VEC

- BSc in Economics of Financial Markets and Institutions (1:1) Full-time: Oct 2009 – Oct 2012
- University of Rome - Tor Vergata Italy & Spain

- Dissertation: The Tobin Tax between Theoretical Wherewithal and Practical Feasibility

TEACHING & MENTORSHIP

Course Instruction & Guest Lectures

- **Imperial College London** – COMP70076 Ethics, Fairness and Explanation in AI (Guest Lecture, Spring 2026): *Causal Discovery for Trustworthy AI*
- **Imperial College London** – COMP70050 Introduction to Machine Learning (Tutorials and Marking, 2021–2022)
- **Luiss Guido Carli, Rome** – BSc Data Science In Action (Guest Lecture, Autumn 2025): *Causal Discovery for Trustworthy AI*
- **Politecnico di Milano & UCL, London** – MSc in Fintech (Guest Lecture, Spring 2022): *From Credit Risk to XAI Research*

Thesis Supervision

- Sam Waugh (MSc Computing 2026): Causal ABA Learning
- Tiancheng Yang (MSc Computing 2026): Explainable AI for Answer Sets and its application to Causal Discovery
- Dhruv Himatsingka (MEng Computing 2026): Bridging Gradual Semantics and GNNs for Argumentative Causal Discovery
- Zihao Li (MSc Computing 2025): Leveraging Large Language Models for Causal Discovery [9]
- Yeva Hunanyan (MSc Computing 2025): Exploring Semantics for Argumentative Causal Discovery
- Preesha Gehlot (MEng Computing 2025): Graph Neural Networks for Approximate Argumentative Reasoning [8]

PROFESSIONAL SERVICE

Invited Research Talks & Seminars

- *Causal AI You Can Challenge: From Data to Decisions in Banking*. **Lloyds Banking Group Summer School**, London UK (2026)
- *CausAlly: Collaborative Causal Discovery for AI-Aided Decision-Making*. **Minds and Machines Workshop, University of Queensland**, Brisbane Australia (2025)
- *Causal Discovery for Trustworthy AI*. **Lloyds Banking Group Summer School**, London UK (2025)
- *Causal Discovery for Contestable and Explainable AI: From Contestable Neural Networks to Transparent Causal Discovery*. **Robert Gordon University**, Aberdeen UK (2024)
- *Credit Risk Modelling: Data and Techniques Used in the UK Banking Industry*. **Danmarks Nationalbank Conference**, Copenhagen Denmark (2019)

Conference & Journal Reviewing

- **Artificial Intelligence Journal**: Reviewer (2026, 2025)
- **KR**: Reviewer (2026)
- **IJCAI-ECAL**: Reviewer (2026, 2023)
- **AAAI**: Reviewer (2026)
- **NeurIPS**: Reviewer (2025, 2024)
- **ICLR**: Reviewer (2025)
- **AISTATS**: Reviewer (2024, 2022); Top-Reviewer (2023)
- **ICML**: Reviewer (2022)

Academic Event Organisation

- **XAI Seminar Series @ Imperial**: Co-organiser (2021–present). Grown from 40 to 450+ subscribers.
- **XAI-FIN at ICAIF**: Co-Chair (2025); Assistant Chair (2023, 2024); Workshop co-organiser (2022). 100+ attendees.

PUBLICATIONS

- [9] Li Z, Russo F. [Leveraging Large Language Models for Causal Discovery: a Constraint-based, Argumentation-driven Approach](#). Proc. of the 42nd Conference on Uncertainty in Artificial Intelligence. pp. 3631–3667. 2026 (CORE: A).
- [8] Gehlot P, Rapberger A, Russo F, Toni F. [Heterogeneous Graph Neural Networks for Assumption-Based Argumentation](#). Proc. of the 40th AAAI Conference on Artificial Intelligence. pp. 19117–19125. 2026 (CORE: A*).
- [7] Rapberger A, Russo F, Rago A, Toni F. [On Gradual Semantics for Assumption-Based Argumentation](#). Proc. of the 22nd International Conference on Principles of Knowledge Representation and Reasoning. pp. 512–522. 2025 (CORE: A*).
- [6] Russo F, Toni F. [Shapley-PC: Constraint-based Causal Structure Learning with a Shapley Inspired Framework](#). Proc. of the 4th Conference on Causal Learning and Reasoning. pp. 292–339. 2025.
- [5] Russo F, Rapberger A, Toni F. [Argumentative Causal Discovery](#). Proc. of the 21st International Conference on Principles of Knowledge Representation and Reasoning. pp. 938–949. 2024 (CORE: A*).
- [4] Leofante F, Ayoobi H, Dejl A, Freedman G, Gorur D, Jiang J, Paulino-Passos G, Rago A, Rapberger A, Russo F, Yin X, Zhang D, Toni F. [Contestable AI Needs Computational Argumentation](#). Proc. of the 21st International Conference on Principles of Knowledge Representation and Reasoning. pp. 888–896. 2024 (CORE: A*).
- [3] Russo F, Toni F. [Causal Discovery and Knowledge Injection for Contestable Neural Networks](#). Proc. of the 26th European Conference on Artificial Intelligence. pp. 2025–2032. 2023 (CORE: A).
- [2] Russo F. [Argumentation for Interactive Causal Discovery](#). Proc. of the 32nd International Joint Conference on Artificial Intelligence. pp. 7091–7092. 2023 (CORE: A*).
- [1] Rago A, Russo F, Albin E, Toni F, Baroni P. [Explaining Classifiers' Outputs with Causal Models and Argumentation](#). Journal of Applied Logics – IfCoLog Journal of Logics and their Applications. Vol. 10, No. 3. pp. 421–449. 2023.

RESEARCH FUNDING & AWARDS

- **Co-Investigator** | *Hypofractionated versus conventional radiotherapy for older adults with glioblastoma: a target trial emulation of survival, hospital burden, and cost-effectiveness* | **NIHR Health Technology Assessment (HTA)** | £600,000 Under review
- **Project Lead** | *CausalABALearn: Learning Non-Monotonic Causal Mechanisms* | **Royal Society International Exchanges (CNR)** | £12,000 Under review
- **Project Lead** | *Research Fellowship: AI in Policy* | **Imperial Policy Forum** | 2 years Awarded
- **Co-Investigator** | *CaRLA: Causal Argumentative Learning Assistant* | **ERC Proof of Concept** | €150,000 Awarded